

Section 2: Amplifier Stages

2026年2月20日 星期五 14:28

An opamp should have a very high voltage gain, which we can achieve with a common emitter amplifier.

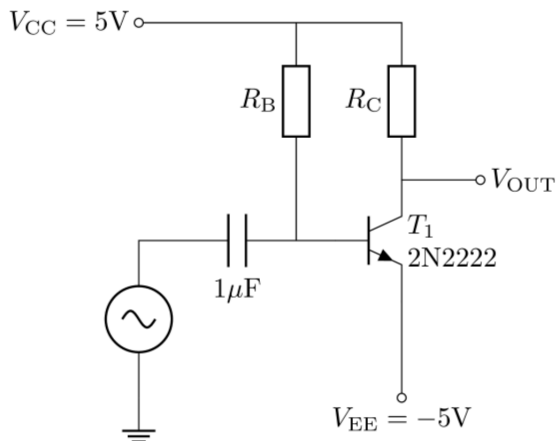


Fig1. CE Amplifier

The base of the transistor is biased from V_{CC} via R_B — select a value that will result in a collector current of 1mA. The capacitor blocks any bias current from the input. R_C should be selected to set V_{OUT} at 0V under quiescent conditions.

Choose bias resistors for the common emitter amplifier and confirm with a DC operating point simulation

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standard.bjt
* Copyright © 2000 Linear Technology Corporation. All rights reserved.
*
*
.model 2N2222 NPN (IS=1E-14 VAF=100
+ BF=200 IKF=0.3 XTB=1.5 BR=3
+ CJC=8E-12 CJE=25E-12 TR=100E-9 TF=400E-12
+ ITF=1 VTF=2 XTF=3 RB=10 RC=.3 RE=.2 Vceo=30 Icrating=800m mfg=NXP)

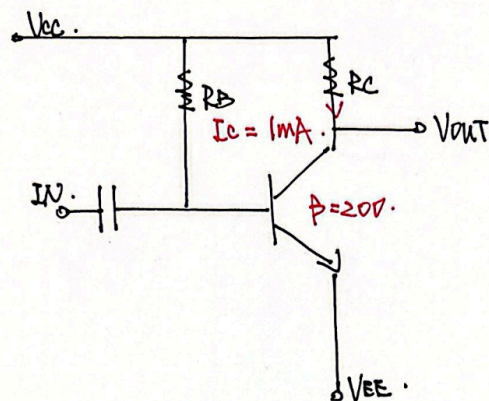
.model 2N2907 PNP (IS=1E-14 VAF=120
+ BF=250 IKF=0.3 XTB=1.5 BR=3
+ CJC=8E-12 CJE=30E-12 TR=100E-9 TF=400E-12
+ ITF=1 VTF=2 XTF=3 RB=10 RC=.3 RE=.2 Vceo=40 Icrating=600m mfg=NXP)

.model 2N3904 NPN (IS=1E-14 VAF=100
+ BF=300 IKF=0.4 XTB=1.5 BR=4
+ CJC=4E-12 CJE=8E-12 RB=20 RC=0.1 RE=0.1
+ TR=250E-9 TF=350E-12 ITF=1 VTF=2 XTF=3 Vceo=40 Icrating=200m mfg=NXP)

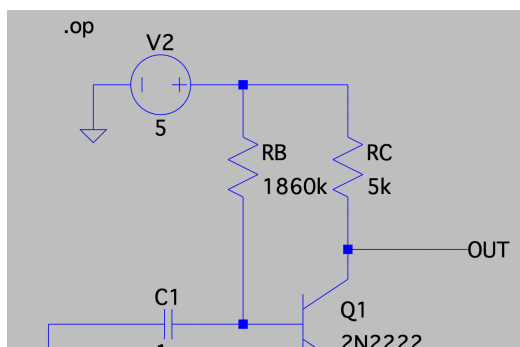
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+ CJC=4E-12 CJE=1E-12 RB=20 RC=0.1 RE=0.1
+ TR=250E-9 TF=350E-12 ITF=1 VTF=2 XTF=3 Vceo=40 Icrating=200m mfg=NXP)

.model FZT849 NPN (IS=5.8591E-13 NF=0.9919 BF=230 IKF=18 VAF=90
+ CJC=2.0635E-12 CJE=1.41E-12 TR=100E-9 TF=400E-12
+ ITF=1 VTF=2 XTF=3 RB=10 RC=.3 RE=.2 Vceo=30 Icrating=800m mfg=NXP)
    
```

Fig2. beta for 2N2222



$$\begin{aligned}
 \textcircled{1} \quad & V_{CC} - V_B - V_{BE} = V_{EE} \\
 & 5 - R_B \cdot \frac{10^{-3}}{200} - 0.7 = -5 \\
 & \therefore R_B = 1860k\Omega \\
 \textcircled{2} \quad & V_{CC} - V_C = 0 \\
 & 5 - 10^{-3} R_C = 0 \\
 & R_C = 5k\Omega
 \end{aligned}$$



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Draft3.log
* Copyright © 2000 Linear Technology Corporation. All rights reserved.
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*
.model 2N2222 NPN (IS=1E-14 VAF=100
+ BF=200 IKF=0.3 XTB=1.5 BR=3
+ CJC=8E-12 CJE=25E-12 TR=100E-9 TF=400E-12
+ ITF=1 VTF=2 XTF=3 RB=10 RC=.3 RE=.2 Vceo=30 Icrating=800m mfg=NXP)

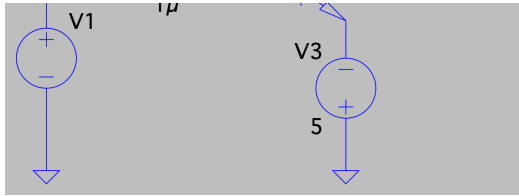
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+ BF=250 IKF=0.3 XTB=1.5 BR=3
+ CJC=8E-12 CJE=30E-12 TR=100E-9 TF=400E-12
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.model 2N3906 PNP (IS=1E-14 VAF=100
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.model FZT849 NPN (IS=5.8591E-13 NF=0.9919 BF=230 IKF=18 VAF=90
+ CJC=2.0635E-12 CJE=1.41E-12 TR=100E-9 TF=400E-12
+ ITF=1 VTF=2 XTF=3 RB=10 RC=.3 RE=.2 Vceo=30 Icrating=800m mfg=NXP)

Operating Bias Point Solution:
V(n001) 5 voltage
V(n003) -4.3445 voltage
V(out) -0.214021 voltage
V(n004) -5 voltage
V(n002) 0 voltage
Ic(Q1) 0.0010428 device_current
Ib(Q1) 5.02392e-06 device_current
Ie(Q1) -0.00104783 device_current
I(C1) -4.3445e-18 device_current
I(Rb) 5.02392e-06 device_current
I(Rc) 0.0010428 device_current
I(V1) -4.3445e-18 device_current
I(V2) -0.00104783 device_current
I(V3) -0.00104783 device_current
    
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Total elapsed time: 0.039 seconds.

Fig3. Simulation

True

ain of the amplifier can be analysed using the small signal equivalent circuit:

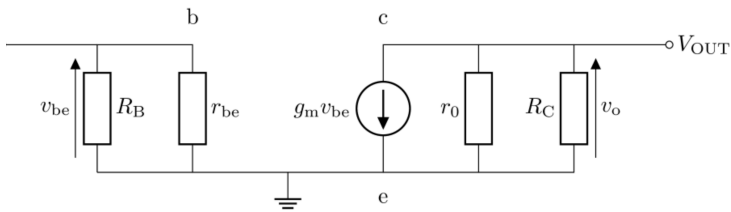


Fig4. SSEM

ut resistance of the transistor:

$$r_{be} = \frac{\beta}{g_m}$$

ll signal output impedance:

$$r_o \approx \frac{V_A}{I_C}$$

all signal transconductance of the transistor which relates the change in collector output current to the change in base-emitter voltage:

$$g_m = \frac{dI_C}{dV_{BE}} = \frac{I_C}{V_T}$$

1 be derived from the equivalent circuit:

$$A_v = \frac{v_o}{v_{in}} = -g_m \left(\frac{R_C r_o}{R_C + r_o} \right)$$

e the theoretical gain of the amplifier

$$\begin{aligned} r_{be} &= \frac{\beta}{g_m} = \frac{\beta}{\frac{I_C}{V_T}} = \frac{200}{\frac{10^{-3}}{0.025}} = 5000 \Omega \\ r_o &= \frac{V_A}{I_C} = \frac{100}{10^{-3}} = 10^5 \Omega \\ \therefore A_v &= -g_m \left(\frac{R_C r_o}{R_C + r_o} \right) \\ &= -190.48 \end{aligned}$$

ting, you will also see that the gain rolls off at low and high frequency. The low frequency used by the input coupling capacitor and the high frequency rolloff is caused by the internal capacitance of the transistor.

orm an AC analysis to find the gain of the circuit.

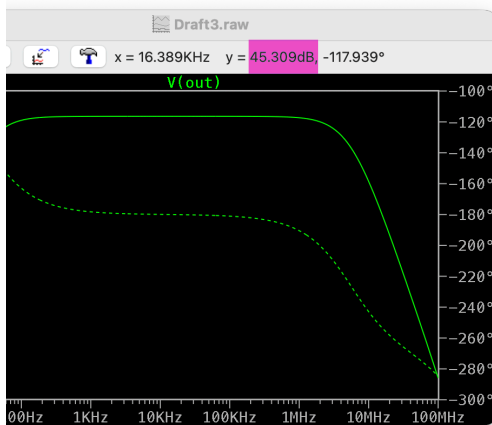


Fig5. AC Analysis of Amplifier

39dB = -178.6.

input resistance

Voltage amplifiers would have infinite input resistance, to ensure that all of the signal voltage from the source is transferred to the amplifier. The input resistance given by the small signal model for our amplifier is:

$$r_{in} = \frac{R_B r_{in}}{R_B + r_{in}}$$

Find r_{in} in simulation by finding the effect of adding a series resistor to the input. If the amplifier did have infinite input resistance resistor would have no effect because there would be no current flowing through it. Add a 10kΩ resistor in series with the input and run the simulation again. You will find the gain has reduced — call the new gain A'_V (in V/V , not dB). You can now calculate the input resistance of the amplifier as:

$$A'_V = A_V \frac{r_{in}}{r_{in} + 10k}$$

$$r_{in} = 10k \frac{A_V}{A_V - A'_V}$$

Verify the input resistance according to the small signal model and verify it with a simulation

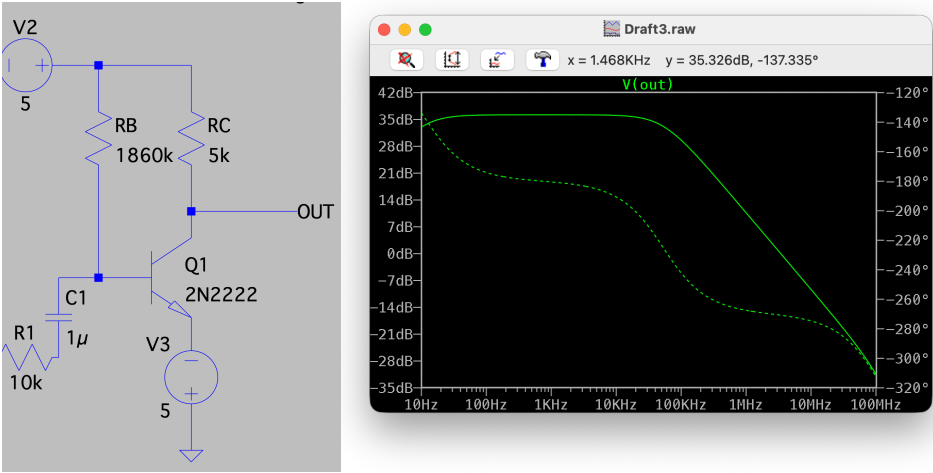


Fig6. Adding a 10k Resistance in Series

35.326dB = 58.385,
the experimental value of $r_{in} = 10k * (58.385 / (184.3 - 58.385)) = 4636.9ohm$.
theoretical value = 4986.6ohm, which is very acceptable.

using an active load

Replace RC with a current mirror, which in this application acts as an *active load*:

A current mirror has a high output resistance and the output current can be set independently of the output voltage by choosing the input current.

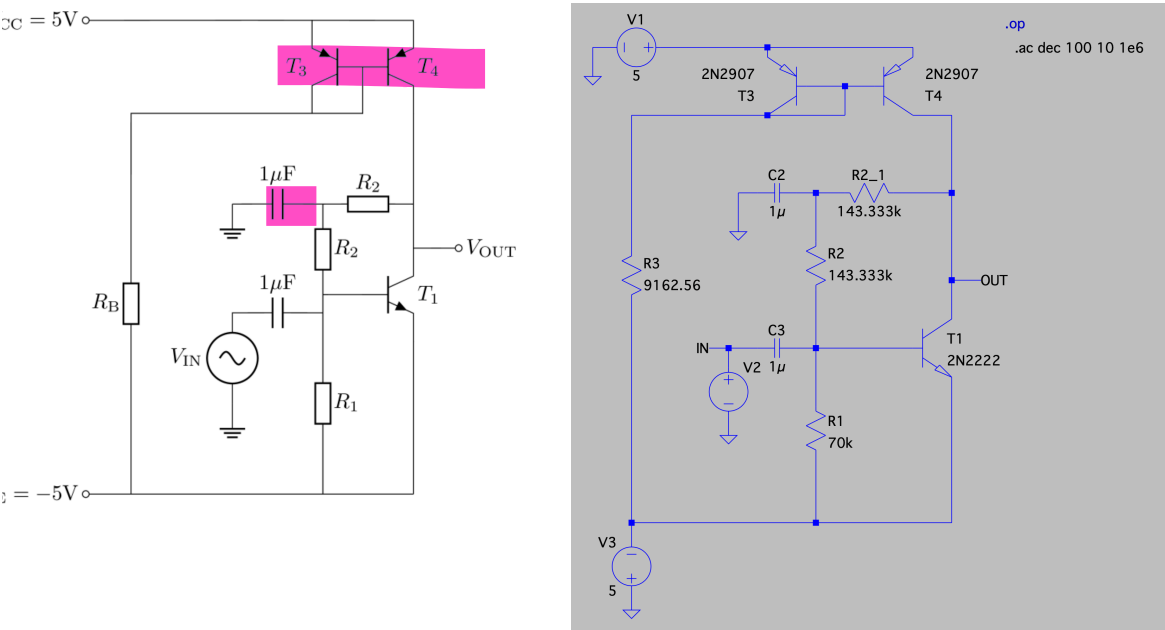


Fig7. Using Active Load

ase current for T_1 is now biased by resistors R_1 and R_2 . A bypass capacitor to ground prevents the bias resistors reducing the AC gain through negative feedback. I_C will be set at 1mA by R_B via the current mirror. Choose R_1 so the current in R_1 is around $2I_B$. Then choose R_2 to set V_{OUT} at 0V, observing that $V_{CB} = 2R_2 \times 3I_B$. Confirm the ig with an operating point simulation and adjust the resistor values if necessary.

that R_C is the output resistance of T_4 , the gain of our amplifier becomes:

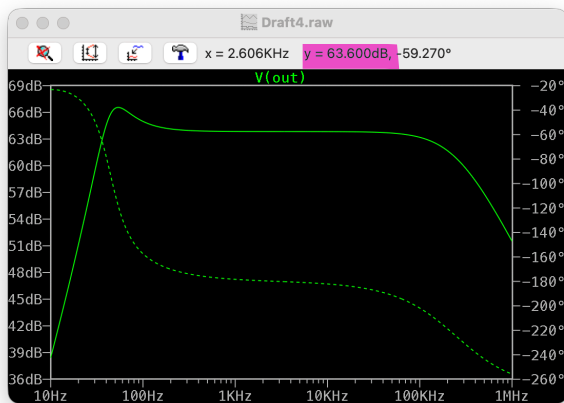
$$A_V = \frac{v_o}{v_{in}} = -g_m \left(\frac{1}{r_{o,T1}} + \frac{1}{r_{o,T4}} + \frac{1}{R_2} \right)^{-1}$$

Add the current mirror to your CE amplifier, set up the biasing and measure the gain with an AC analysis

With the active load, the effective output resistance is the parallel combination of r_o of T_1 (NPN) and r_o of T_4 (PNP mirror):

$$\begin{aligned} A_V &= -g_m \cdot \left(\frac{1}{r_{o,T1}} + \frac{1}{r_{o,T4}} + \frac{1}{R_2} \right)^{-1} \\ &= -0.04 \times \left(\frac{I_0}{V_A} + \frac{I_0}{2 \cdot V_A} + \frac{1}{143.33k} \right)^{-1} \\ &= -0.04 \times \left(\frac{1m}{100} + \frac{2 \times 10^{-3}}{2 \times 100} + \frac{1}{143.33k} \right)^{-1} \\ &\approx -1482.8 \Rightarrow 63.42dB \end{aligned}$$

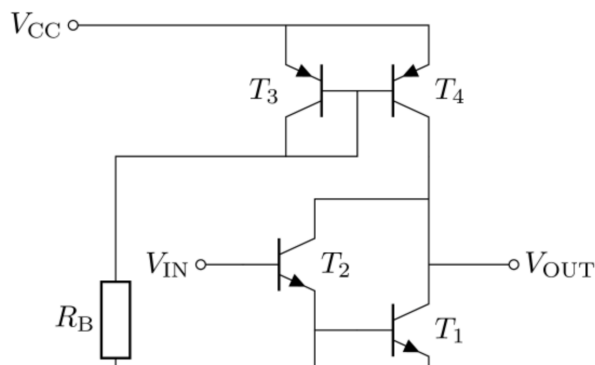
The measured midband gain from the AC simulation was 63.6 dB, in excellent agreement with the theoretical value of 63.42 dB.



Acceptable.

Darlington Pair

Earlier you found the [input resistance](#) of the common emitter amplifier, which wasn't very high. We can improve it using a Darlington pair configuration:





Since I_B for T_1 comes from the emitter of T_2 , the input current for the amplifier is decreased by a factor of approximately β . The resistor R_E is used to make sure that the collector bias current is split equally between T_1 and T_2 . This ensures that both transistors have similar high frequency behaviour (based on capacitance), which is dependent on I_C .

Including the effect of R_E , the input resistance is now:

$$r_{in} = \beta \left(\frac{1}{R_E} + \frac{g_m}{\beta} \right)^{-1}$$

It is easier to bias this amplifier when it is integrated with the other amplifier blocks, so you don't need to simulate it on its own.